

BONUS E-LECTURE:

Allergy & Clinical Immunology

Prepared by Dr. Adhora Mir for IMR 2025



Overview

1. Anaphylaxis

2. Cutaneous Disorders

- Urticaria
- Angioedema

3. Drug allergy

- Penicillin Allergy
- “Pseudoallergic” reactions
- Drug rashes

4. Immunodeficiency

- CVID

5. Bonus Slides

- Primary Immunodeficiency
- Mastocytosis
- Eosinophilia
- Common myths
- Bonus MCQs

*Please see GI lecture for **Eosinophilic Esophagitis**

*Please see Respiriology lecture for **Asthma**

Anaphylaxis

Diagnosis

Recent Exposure  None	Skin and/or Mucosa Pruritus, flushing, hives, angioedema + Respiratory Dyspnea, stridor, hypoxemia, wheeze or ↓BP or End Organ Damage Syncope, incontinence
Recent Exposure  Likely	≥2 of: Skin and/or Mucosa Pruritus, flushing, hives, angioedema Respiratory Dyspnea, stridor, hypoxemia, wheeze Persistent GI symptoms Vomiting, diarrhea, cramping abdominal pain ↓BP or End Organ Damage Syncope, incontinence
Recent Exposure  Known	↓BP

Adapted from NIAD/FAAN 2006

Ddx: Vasovagal, flushing, vocal cord dysfunction, panic attack, scombroid (spoiled fish), angioedema

- **Main mechanism:** IgE binding, cross-linking causing mast cell degranulation
- **Triggers:** Foods, medications, insect stings, radiocontrast media, perioperative agents, exercise, seminal fluid, idiopathic
- **Management:**
- **Stabilize:** ABCs, discontinue trigger if known, supine position
- **Epinephrine** (IM > IV; lateral thigh > arm)
 - **IM: 0.01 mg/kg (max 0.5mg)** IM Repeat q5-15 min PRN (1:1000 solution = 1mg/mL)
 - **IV: 0.05-0.1 mg IV over 5 min**, infuse 2-10 mcg/min titrated to BP (1:10,000 solution = 0.1mg/mL)
- **Glucagon** if taking beta blockers
- **Supportive:** Inhaled SABA, antihistamines, supplemental O2, IV crystalloid fluids, vasopressors, corticosteroids
- **Refractory:** IV epinephrine, intubation, methylene blue
- **Monitor:** Transfer to hospital ED, monitor for at least an hour after symptoms resolve; Tryptase x1 (peaks in 1hr, useful for consultant)
- **Non pharm:** Educate on avoidance, recognition, and action plan, Rx Epi-pen and teach technique, MedicAlert bracelet or allergist consultation if unknown trigger

Cutaneous Disorders

Urticaria

- **Presentation:** Transient (<48hrs), pruritic, erythematous wheals of variable sizes
 - Acute (<6 weeks) or chronic ($\geq$ 6 weeks)
- **Common triggers (obvious relationship):** infections, food, drugs, venom, physical (cold, dermatographism, delayed pressure, solar, water, contact)
- **Co-factors:** Stress, NSAIDs, hormone therapy/menstruation, exercise, EtOH
- **Red flag symptoms:** lesions >48h, non-blanching, painful, scarring, systemic symptoms $\rightarrow$ urticarial vasculitis, autoinflammatory syndrome, other dermatosis

Chronic Spontaneous Urticaria (CSU)

- Hives occurring most days of the week >6 weeks, no clear trigger +/- angioedema (rarely only angioedema)
- Often associated with autoimmunity
- Investigations (depending on history)
 - Autoimmune: CBC + diff, ESR/CRP, TSH/TPO antibody
 - Mastocytosis: tryptase
 - Unresolving: biopsy, total IgE

Management:

Daily non-sedating antihistamine (cetirizine)
 $\downarrow$
Increase non-sedating antihistamine dose up to 4x
 $\downarrow$
Omalizumab (Anti-IgE)
 $\downarrow$
Cyclosporin

Angioedema

Histamine Mediated

- Pruritic, 24-48hr, often + hives, may have associated systemic changes, and responds to epinephrine/antihistamines.
- **Anaphylaxis:** Systemic, typically within 1 hour
 - Secure airway, epinephrine, remove trigger
- **Acute allergen triggered**
- **CSU associated angioedema**
- **Pseudoallergic angioedema:** Typically associated with other drug classes, such as NSAIDs.

Bradykinin mediated:

- Generally non pruritic/burning, asymmetric nondependent swelling, days to resolve



- **ACE inhibitor induced**
 - Risk: African American, smoking, age, female → **Stop ACEi**
- **Acquired C1INH deficiency** (Increased C1INH catabolism)
 - Risk: Middle aged or older adults, autoimmune or lymphoproliferative disease → treat underlying disease
- **Hereditary (HAE):** Typically childhood, FHx, recurrent abdominal pain. Mutation resulting in low levels or impaired C1INH.
 - **Investigations:** C4 (low in acute), C1INH level/function, genetic studies
 - **Management:**
 - Education, genetic counseling, Medic Alert bracelet
 - C1INH, Icatibant (bradykinin receptor antagonist)

Drug Allergy

Penicillin Allergy

Most common drug allergy, however >90% labels inaccurate!

If IgE reaction suspected (immediate, symptoms of anaphylaxis)

- Allergy referral for skin testing +/- oral challenge
- For adults with history of distant (>5 years) and benign cutaneous reaction, direct amoxicillin challenge should be considered (2022 AAAAI Drug

Allergy Practice Parameter Update)

If delayed hypersensitivity reaction suspected

- Clinical Diagnosis remains gold standard
- Continue to avoid in the case of severe cutaneous adverse reactions (e.g. DRESS, SJS/TEN)

Desensitization may be considered if penicillin is required acutely and no time for skin testing, or skin test positive

- Note: this is a temporary induction of 'tolerance', and does NOT rule out allergy
- Note: this is only used in cases of IgE-mediated drug reactions (NOT if any suspicion of serum sickness, SJS-TENS, DRESS, etc.)

Beta-lactam Cross-Reactivity

Dependent on structural similarity, and is low in most cases. For example, cefazolin, ceftriaxone, and carbapenams all have dissimilar R1 side chains and have rates of adverse reactions similar to the general public.

“Pseudoallergic” Reactions

Immediate type non IgE mediated reactions can present through direct mast cell/basophil activation. Commonly cutaneous, dose dependent. If multisystem involvement → **think Anaphylaxis, give epi!**

Opioids

- Decrease dose, alternating, premedicate with antihistamines

Radiocontrast

- Hydration, use a nonionic, iso-osmolar RCM, especially in high-risk patients (previous reaction, asthmatics, taking B-blockers, CV disease)
- Consider pretreatment regimen with steroids, antihistamines

Infusions

- E.g. Rituximab, vancomycin
- Antihistamines +/- steroids, resume at slower rate

N-acetylcysteine

- Limited cutaneous symptoms → treat with AH
- Any resp symptoms, angioedema, low BP → Stop infusion +/- epi
- Restart 1hr after symptom resolution if needed
 - PO NAC if hypotensive

Drug Rashes

Common culprits: antibiotics (vancomycin, sulfa, B-lactam), AEDs, allopurinol, NSAIDs.

Morbilliform Drug Eruption

- Most common, pruritic macules/papules on trunk/proximal extremities (mimics viral exanthem)
- **Timeline:** Onset 72h-10d after drug, fades after 1-3 weeks.
- **Treatment:** Symptomatic treatment anti-H2, topical corticosteroid; consider stop drug



Drug Reaction with Eosinophilia and Systemic Symptoms

- Morbilliform rash +/- exfoliative/pustules, facial edema, fever, eosinophilia, lymphadenopathy, organ involvement (most commonly hepatitis)
 - RegiSCAR score for diagnosis/prognosis (mortality $\leq 10\%$)
- **Timeline:** Presents 2-8 weeks after drug, lasts >15 days
- **Treatment:** Oral corticosteroids with slow taper +/- IVIG; AVOID drug



Stevens-Johnson Syndrome/ Toxic Epidermal Necrolysis

- Prodromal URTI symptoms, bullae, mucous membrane ulcerations, conjunctivitis, skin detachment (mortality 5-40%)
 - Differentiated by epidermal detachment %BSA (<10% SJS vs. $\geq 30\%$ TEN)
- **Timeline:** Typically 1-4 weeks after drug, lasts weeks - months
- **Treatment:** Supportive treatment in ICU/burn unit, treat with corticosteroids/ cyclosporine/ infliximab/ etanercept (multidisciplinary decision), avoid drug



Images: AAAAI Mar 2024

Immunodeficiency

Common Variable Immunodeficiency (CVID)

- Most common Primary Immunodeficiency
 - **Presentation:** sinopulmonary infections, granulomatous disease, autoimmunity (ITP, AIHA), malignancy, diarrhea
 - **Diagnosis:**
LOW IgG + LOW IgA or IgM + poor response to vaccination + other causes ruled out
 - **Treatment:** IVIG or SCIG (subcutaneous Immunoglobulin)

Hypogammaglobulinemia DDx:

Drug related (immunosuppressants, steroids, AEDs), lymphoproliferative malignancy, protein loss (nephrotic syndrome, malabsorption, burns) other PID or genetic syndromes

IVIG Risk Counseling

- Mild, common: headache, fever, chills, and fatigue (typically resolve with reduced rate/dose)
- Severe, uncommon: Fluid overload, hemolytic anemia, aseptic meningitis, hepatitis, dyspnea, anaphylaxis

Bonus Slides

Primary Immunodeficiency (PID)

Ten Red Flags of Immunodeficiency:

1. ≥ 2 new ear infections within 1 year
2. ≥ 2 new sinus infections within 1 year, in the absence of allergy
3. One pneumonia per year for more than 1 year
4. Chronic diarrhea with weight loss
5. Recurrent viral infections (colds, herpes, warts, condyloma)
6. Recurrent need for IV antibiotics to clear infections
7. Recurrent, deep abscesses of the skin or internal organs
8. Persistent thrush or fungal infection on skin or elsewhere
9. Infection with normally harmless tuberculosis-like bacteria
10. A family history of Primary Immunodeficiency

Jeffery Modell Foundation

Approach to PID

Secondary causes of immunodeficiency:

- DM, HIV infection, Cirrhosis, Nephrotic syndrome, Autoimmune disease, Malignancy, Splenectomy/Asplenia, Immunomodulatory drugs
- Structural (obstructive tumours, urethral strictures), dermatitis, burns

Investigative Approach

Type of Infection	Type of Immunodeficiency	Investigation
Repeated pyogenic infections	B-Cell	Lymphocyte count, Lymphocyte subsets, Immunoglobulins (IgG, IgA, IgM), vaccination titres
Severe mycotic infection and opportunistic infections	T-Cell	Lymphocyte count, Lymphocyte subsets
Abscess-forming infection with low-grade pathogens	Neutrophil deficiency	Neutrophil count, Chronic Granulomatous Disease (CGD) Assay
Repeated infections w/ <i>Nisseria</i> sp.	Complement Deficiency	C3, C4, CH50

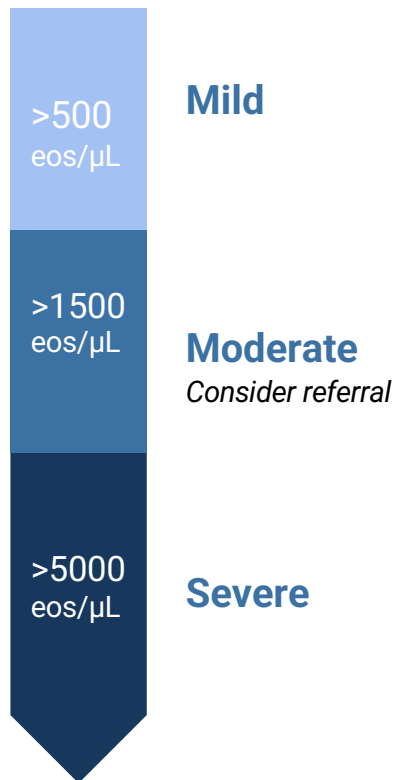
Mastocytosis

- Excessive amount of mast cells in one or more tissues (may be tumours, cutaneous, or systemic)
- Adult form associated with c-KIT mutation in 90% of cases
- **Cutaneous:**
 - Urticaria pigmentosa (urticaria with trauma=Darier's sign); Telangiectasia macularis eruptiva perstans, diffuse cutaneous (may have bullae/hemorrhage)
- **Systemic**
 - **Diagnostic criteria:** 1 major + 1 minor or 3 minor
 - **Major:** Multifocal dense mast cell infiltrates
 - **Minor:** typical appearance of infiltrate, c-KIT mutation, CD2/CD25 expression on CD117+ mast cell, serum tryptase >20 without other hematologic disorder
 - **Presentation:** Constitutional, severe allergic reactions, headaches, GI, typical skin findings/pruritus/flushing, bone pain/lesions, uncommonly cardiorespiratory symptoms
 - **Treatment:** antihistamines, mast cell stabilizers, phototherapy, omalizumab, cytoreductive agents

Eosinophilia

Differential Diagnosis:

- **Neoplasms**
 - Myeloid and lymphocytic (Ix= FISH Panel, B12, tryptase, splenomegaly, chronic eosinophilic leukemia on BM/smear)
- **Allergic disease**
 - Typically mild with single system involvement, drug-related can be more severe
- **Immune Dysregulation**
 - Eosinophilic granulomatosis with polyangiitis or other autoimmune disease
- **Infections**
 - Parasites, fungal, HIV (Ix= Stool O&P x 3, strongyloides serology)
- **Hypereosinophilic syndrome**
 - Sustained eosinophils >1500 cells/uL x2 separated by at least 1 mo + Evidence of end-organ involvement + Exclusion of other disorders



Common Myths

- ✓ Radiocontrast reactions are not related to shellfish allergy
- ✓ Those with egg allergy can safely receive most vaccinations
- ✓ Generally, only those IgA deficiency patients with undetectable levels of IgA are at risk for anaphylaxis with blood products

Bonus MCQ 1

A 30-year-old woman presents with a second pneumonia in the last six months. She has a past medical history of recurrent pneumonias and sinusitis. Investigations show: WBC 6, Neutrophils 3, Lymphocytes 2, Monocytes 1.

What is most likely diagnosis?

1. Complement deficiency
2. Common variable immunodeficiency
3. HIV
4. Chronic granulomatous disease

Answer

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What is most likely diagnosis?

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3. HIV
4. Chronic granulomatous disease

Bonus MCQ 2

A middle-aged female presents with a 6 month history of urticaria. She has never had any episodes of swelling. The urticaria resolve within 12hr, with no residual bruising or scarring. She reports that she has been eating out at restaurants more recently. She is otherwise well. Initial bloodwork is normal including a CBC, creatinine and liver profile. What is the most likely cause of this patient's symptoms?

1. Chronic Idiopathic Urticaria
2. Food allergy
3. C1 esterase deficiency
4. Autoimmune Disease

Answer

A middle-aged female presents with a 6 month history of urticaria. She has never had any episodes of swelling. The urticaria resolve within 12hr, with no residual bruising or scarring. She reports that she has been eating out at restaurants more recently. She is otherwise well. Initial bloodwork is normal including a CBC, creatinine and liver profile. What is the most likely cause of this patient's symptoms?

- 1. Chronic Idiopathic Urticaria**
2. Food allergy
3. C1 esterase deficiency
4. Autoimmune Disease

Bonus MCQ 3

A pregnant patient with a history of asthma states she is allergic to penicillin. She developed wheeze the last time she took it, while admitted for an upper respiratory tract infection. She was referred for penicillin skin testing, which was negative.

Penicillin is now indicated for a pregnancy acquired syphilis infection. Which drug would you use?

1. Penicillin
2. Second-Generation Cephalosporin
3. Doxycycline

Answer

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Penicillin is now indicated for a pregnancy acquired syphilis infection. Which drug would you use?

- 1. Penicillin**
2. Second-Generation Cephalosporin
3. Doxycycline

Bonus MCQ 4

A young man presents with occasional angioedema of his lips and hands. Which of the following features make it least consistent with C1 esterase deficiency?

1. Occasional GI symptoms
2. Triggered by dental procedure
3. Presence of urticaria
4. Laryngeal involvement

Answer

A young man presents with occasional angioedema of his lips and hands. Which of the following features make it **least** consistent with C1 esterase deficiency?

1. Occasional GI symptoms
2. Triggered by dental procedure
- 3. Presence of urticaria**
4. Laryngeal involvement

Bonus MCQ 5

A 68 year old male presents with undetectably high acetaminophen level and is started on N-acetylcysteine (NAC) infusion. Two hours into the infusion, he develops urticaria and hypotension. What are the next most appropriate steps?

1. Stop infusion, give IM epinephrine, restart IV NAC
2. Stop infusion, give antihistamines, restart IV NAC
3. Stop infusion, give IM epinephrine, begin PO NAC
4. Stop infusion, give antihistamines, avoid NAC

Answer

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4. Stop infusion, give antihistamines, avoid NAC

Bonus MCQ 6

A 60 year old female known for hypertension, coronary artery disease, and type 2 diabetes on chlorthalidone, metoprolol and metformin has anaphylaxis to pre-operative cefazolin. IM epinephrine is administered twice with minimal effect. What adjunctive therapy would be most effective for this patient?

1. Benadryl
2. Glucagon
3. Methyl prednisone
4. Omalizumab

Answer

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2. **Glucagon**
3. Methyl prednisone
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Bonus MCQ 7

A patient with history of UTI 2 months ago treated with TMP-SMX presents with worsening maculopapular rash over trunk and extremities over the last week. He presents to the ED with new fever and lymphadenopathy, and labs are as follows: WBC 13, eos 3.7, Cr 250, ALT 150, AST 200. How would you treat this patient?

1. Observe
2. IM epinephrine
3. Topical corticosteroids
4. Prednisone

Answer

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Thank You!